fn sample_discrete_laplace

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_discrete_laplace` in `mod.rs` at commit 0be3ab3e6 (up to date). This proof is an adaptation of subsection 5.2 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\texttt{scale} \in \mathbb{Q} \wedge \texttt{scale} > 0$

Implementation

```
def sample_discrete_laplace(scale) -> int:
    if (scale == 0):
        return 0

    inv_scale = recip(scale)

    while True:
        sign = sample_standard_bernoulli()
        magnitude = sample_geometric_exp_fast(inv_scale)

        if sign or magnitude != 0:
            if sign:
                return -magnitude
            else:
                return magnitude
```

Postcondition

`sample_discrete_laplace` returns an integer sample from $\mathcal{L}_{\mathbb{Z}}(0, scale)$, or an error propagated from `sample_geometric_exp_` or `sample_standard_bernoulli`.

3 Proof

First, define the target distribution.

Definition 3.1. (Discrete Laplace). Let $\mu, \sigma \in \mathbb{R}$ with $\sigma > 0$. The discrete laplace distribution with location μ and scale s is denoted $\mathcal{L}_{\mathbb{Z}}(\mu, s)$. It is a probability distribution supported on the integers and defined by [BV17]

$$\forall x \in \mathbb{Z} \quad P[X = x] = \frac{e^{-1/s} - 1}{e^{-1/s} + 1} e^{-|x|/s} \quad \text{where } X \sim \mathcal{L}_{\mathbb{Z}}(\mu, s)$$

Lemma 3.2. Let $B \sim \mathit{Bernoulli}(1/2)$ and $Y \sim \mathit{Geometric}(1 - e^{-1/s})$ for some $s > 0$. Then $P[(B, Y) \neq (\top, 0)] = \frac{1}{2}(e^{-1/s} + 1)$[CKS20].

Proof.

$$\begin{aligned}
P[(B, Y) \neq (\top, 0)] &= P[B = \top, Y > 0] + P[B = \bot] && \text{by LOTP} \\
&= P[B = \top]P[Y > 0] + P[B = \bot] && \text{by independence of B, Y} \\
&= \frac{1}{2}e^{-1/s} + \frac{1}{2} \\
&= \frac{1}{2}(e^{-1/s} + 1)
\end{aligned}$$

□

Theorem 3.3. Given random variables $B \sim \mathit{Bernoulli}(1/2)$ and $Y \sim \mathit{Geometric}(1 - e^{-1/s})$, define $X|_{B=\top} = Y$, and $X|_{B=\bot} = -Y$. If $(B, Y) \neq (\top, 0)$, then $X \sim \mathcal{L}_{\mathbb{Z}}(0, \mathit{scale})$. That is, $P[X = x|(B, Y) \neq (\top, 0)] = \frac{1-e^{-1/\sigma}}{1+e^{-1/\sigma}} e^{-|x|/\sigma}$ for any $x \in \mathbb{Z}$[CKS20].

Proof.

$$\begin{aligned}
P[X = x|(B, Y) \neq (\top, 0)] &= \frac{P[X = x, (B, Y) \neq (\top, 0)]}{P[(B, Y) \neq (\top, 0)]} \\
&= \frac{P[X = |x|, B = \mathbb{I}[x < 0]]}{P[(B, Y) \neq (\top, 0)]} && \text{since x} = \pm y \\
&= \frac{P[X = |x|]P[B = \mathbb{I}[x < 0]]}{P[(B, Y) \neq (\top, 0)]} && \text{by independence of B, Y} \\
&= \frac{P[X = |x|]\frac{1}{2}}{\frac{1}{2}(e^{-1/s} + 1)} && \text{by 3.2} \\
&= \frac{1 - e^{-1/s}}{1 + e^{-1/s}} e^{-|x|/s}
\end{aligned}$$

□

Since `sign` is a realization of B, `magnitude` is a realization of $Y \sim \mathit{Geometric}(1 - e^{-1/\mathit{scale}})$, and the return is conditioned on $(B, Y) \neq (\top, 0)$ and not returning an error, then by 3.3 the function's return value is a realization of $\mathcal{L}_{\mathbb{Z}}(0, \mathit{scale})$.

References

[BV17] Victor Balcer and Salil P. Vadhan. Differential privacy on finite computers. *CoRR*, abs/1709.05396, 2017.

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.